MATHS MANTHAN
• TUITION CLASSES •
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Subject: Mathematics **WEEKLY CHAPTER TEST – 02 (SOLUTIONS)** **Time:** 45 Minutes
Class: Class X (CBSE) **Chapter 12: Surface Areas and Volumes** **MM: 20**

Document Type: Official Step-by-Step Marking Scheme **Published for:** mathsmanthan.co.in

CBSE STEP-MARKING GUIDELINES & EVALUATION PROTOCOL:

1. Full marks require explicit formula declaration, factoring common terms like πr before substituting $\pi = \frac{22}{7}$ or 3.14, showing intermediate arithmetic clearly, and boxing the final answer with proper physical units ($\text{cm}^2, \text{cm}^3, \text{m}^2, \text{m}^3$).
2. Partial step marks are allocated in brackets (**[0.5 Mark]**, **[1 Mark]**, **[1.5 Marks]**, **[2 Marks]**). Alternative valid methods receive full credit.

Official Evaluation Key | Maths Manthan Meerut

SECTION A: Multiple Choice Questions & Assertion-Reason MATHS MANTHAN | (4 × 1 = 4 Marks)

Q1. A solid metallic sphere of radius r is melted and cast into the shape of a solid cone of height r. The radius of the base of the cone is: [1]

(A) r (B) $2r$ (C) $3r$ (D) $4r$

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: **(B) 2r** [0.5 Mark]

Given: Radius of sphere $= r$, height of cone $h = r$. Let the base radius of the cone be R.

Formula / Theorem Used: Volume of Sphere $= \frac{4}{3}\pi r^3$ and Volume of Cone $= \frac{1}{3}\pi R^2 h$

Equating both volumes ($h = r$): [0.5 Mark]

$$\frac{1}{3}\pi R^2(r) = \frac{4}{3}\pi r^3 \implies R^2 = 4r^2 \implies R = \sqrt{4r^2} = \boxed{\mathbf{2r}}$$

Q2. A metallic solid cone is melted to form a solid cylinder of equal radius. If the height of the cylinder is 6 cm, then the height of the cone was: [1]

(A) 10 cm (B) 12 cm (C) 18 cm (D) 24 cm

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: **(C) 18 cm** [0.5 Mark]

Given: Equal base radius r; cylinder height $H = 6$ cm. Let cone height be h.

Formula / Theorem Used: Volume of Cone $= \frac{1}{3}\pi r^2 h$ and Volume of Cylinder $= \pi r^2 H$

Equating volumes and cancelling πr^2: [0.5 Mark]

$$\frac{1}{3}\pi r^2 h = \pi r^2(6) \implies \frac{h}{3} = 6 \implies h = 3 \times 6 = \boxed{\mathbf{18\ cm}}$$

Q3. A rectangular sheet of paper 40 cm × 22 cm is rolled along its breadth to form a hollow cylinder of height 40 cm. The radius of the cylinder is: [1]

(A) 3.5 cm (B) 7 cm (C) $\frac{80}{7}$ cm (D) 5 cm

▶ STEP-BY-STEP SOLUTION: [1 Mark]

Correct Option: **(A) 3.5 cm** [0.5 Mark]

Given: Since height $h = 40$ cm, the 22 cm edge forms the base circumference of the cylinder.

Formula / Theorem Used: Circumference of Circular Base $(C) = 2\pi r$

Substituting $C = 22$ cm and $\pi = \frac{22}{7}$: [0.5 Mark]

$$2 \times \frac{22}{7} \times r = 22 \implies \frac{2r}{7} = 1 \implies r = \frac{7}{2} = \boxed{\mathbf{3.5}\text{ cm}}$$

Q4. **Assertion (A):** Two cubes each of surface area 96 cm^2 are joined end-to-end to form a cuboid. The volume of the resulting cuboid is 128 cm^3.

Reason (R): If n cubes each of surface area S are joined end-to-end, the volume of the resulting cuboid is $n\left(\frac{S}{6}\right)^{3/2}$. [1]

(A) Both (A) and (R) are true, and (R) is the correct explanation of (A).

(B) Both (A) and (R) are true, but (R) is not the correct explanation of (A).

(C) (A) is true, but (R) is false. (D) (A) is false, but (R) is true.

▶ STEP-BY-STEP SOLUTION: [1 Mark]

Correct Option: **(A)** [0.5 Mark]

Verification: Each cube has surface area $S = 6a^2 \implies a = \sqrt{\frac{S}{6}} = \left(\frac{S}{6}\right)^{1/2}$. Joining n cubes gives volume $V = na^3 = n\left(\frac{S}{6}\right)^{3/2}$, so (R) is true. [0.5 Mark]

For $n = 2$ and $S = 96$ cm^2: $a = \sqrt{\frac{96}{6}} = \sqrt{16} = 4$ cm, so $V = 2 \times 4^3 = 128$ cm^3. Both (A) and (R) are true, and (R) explains (A).

SECTION B: Short Answer Questions

MATHS MANTHAN | **(2 × 2 = 4 Marks)**

Q5. A solid is in the shape of a cone surmounted on a hemisphere, the radius of each of them being 3.5 cm and the total height of the solid being 9.5 cm. Find the volume of the solid. [2]

▶ STEP-BY-STEP SOLUTION & MARKING SCHEME: [Total: 2 Marks]

Step 1: Finding Height of Cone & Formula Declaration [1 Mark]

Given: Common radius $r = 3.5$ cm $= \frac{7}{2}$ cm; total height $= 9.5$ cm.

Height of cone $h = 9.5 - r = 9.5 - 3.5 = 6$ cm.

Formula / Theorem Used: Volume $= V_{\text{cone}} + V_{\text{hemi}} = \frac{1}{3}\pi r^2 h + \frac{2}{3}\pi r^3 = \frac{1}{3}\pi r^2(h + 2r)$

Step 2: Substitution & Arithmetic [1 Mark]

$$V = \frac{1}{3} \times \frac{22}{7} \times \frac{7}{2} \times \frac{7}{2} \times (6 + 7) = \frac{11 \times 7 \times 13}{6} = \frac{1001}{6} = \boxed{\mathbf{166\frac{5}{6}}\text{ cm}^3\ (\approx \mathbf{166.83}\text{ cm}^3)}$$

Q6. The largest possible sphere is carved out of a wooden solid cube of side 7 cm. Find the volume of the wood left. [2]

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 2 Marks]

Step 1: Dimensions & Formula Declaration [1 Mark]

Given: Cube side $a = 7$ cm. The largest sphere has diameter $= 7$ cm $\implies r = \frac{7}{2}$ cm.

Formula / Theorem Used: Wood Left $= V_{\text{cube}} - V_{\text{sphere}} = a^3 - \frac{4}{3}\pi r^3$

Step 2: Substitution & Subtraction [1 Mark]

$V_{\text{cube}} = 7^3 = 343$ cm^3; $V_{\text{sphere}} = \frac{4}{3} \times \frac{22}{7} \times \frac{7}{2} \times \frac{7}{2} \times \frac{7}{2} = \frac{11 \times 49}{3} = \frac{539}{3}$ cm^3.

$$\text{Wood Left} = 343 - \frac{539}{3} = \frac{1029 - 539}{3} = \frac{490}{3} = \boxed{\mathbf{163\frac{1}{3}}\ \text{cm}^3\ (\approx \mathbf{163.33}\ \text{cm}^3)}$$

SECTION C: Short Answer Questions

MATHS MANTHAN | (2 × 3 = 6 Marks)

Q7. A toy is in the form of a cone of radius 3.5 cm mounted on a hemisphere of the same radius. The total height of the toy is 15.5 cm. Find the total surface area of the toy. [3]

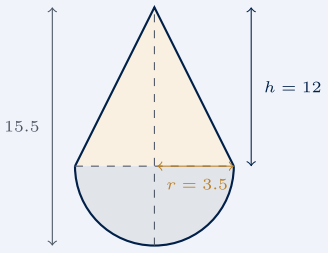
▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 3 Marks]

Step 1: Heights & Slant Height [1 Mark]

Common radius $r = 3.5$ cm $= \frac{7}{2}$ cm.

Height of cone $h = 15.5 - 3.5 = 12$ cm.

Slant height $l = \sqrt{r^2 + h^2} = \sqrt{(3.5)^2 + 12^2} = \sqrt{12.25 + 144} = \sqrt{156.25} = 12.5$ cm.



Step 2: Formula Declaration [1 Mark]

Formula / Theorem Used: TSA of Toy $= \text{CSA}_{\text{cone}} + \text{CSA}_{\text{hemi}} = \pi rl + 2\pi r^2 = \pi r(l + 2r)$

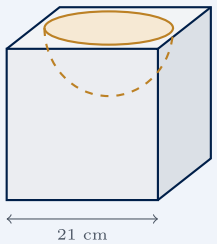
Step 3: Substitution & Answer [1 Mark]

$$\text{TSA} = \frac{22}{7} \times \frac{7}{2} \times (12.5 + 7) = 11 \times 19.5 = \boxed{\mathbf{214.5}\ \text{cm}^2}$$

Q8. A hemispherical depression is cut out from one face of a cubical wooden block of edge 21 cm, such that the diameter of the hemisphere is equal to the edge of the cube. Determine the volume and total surface area of the remaining block. [3]

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 3 Marks]

Given: Cube edge $a = 21$ cm; hemisphere diameter $= 21$ cm $\implies r = \frac{21}{2}$ cm $= 10.5$ cm.



Step 1: Volume of Remaining Block [1.5 Marks]

Formula / Theorem Used: $V_{\text{rem}} = a^3 - \frac{2}{3}\pi r^3$ and $\text{TSA}_{\text{rem}} = 6a^2 - \pi r^2 + 2\pi r^2 = 6a^2 + \pi r^2$

$V_{\text{cube}} = 21^3 = 9261$ cm^3; $V_{\text{hemi}} = \frac{2}{3} \times \frac{22}{7} \times \frac{21}{2} \times \frac{21}{2} \times \frac{21}{2} = \frac{11 \times 441}{2} = 2425.5$ cm^3.

$$V_{\text{rem}} = 9261 - 2425.5 = \boxed{\mathbf{6835.5}\ \text{cm}^3}$$

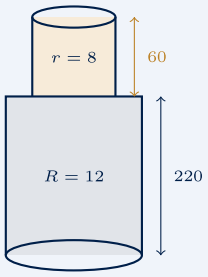
Step 2: Total Surface Area [1.5 Marks]

$6a^2 = 6 \times 441 = 2646$ cm^2; $\pi r^2 = \frac{22}{7} \times \frac{21}{2} \times \frac{21}{2} = \frac{11 \times 63}{2} = 346.5$ cm^2.

$$\text{TSA}_{\text{rem}} = 2646 + 346.5 = \boxed{\mathbf{2992.5}\ \text{cm}^2}$$

Q9. A solid iron pole consists of a cylinder of height 220 cm and base diameter 24 cm, which is surmounted by another cylinder of height 60 cm and radius 8 cm.
(i) Calculate the total volume of the iron pole. (Use $\pi = 3.14$)
(ii) Find the mass of the pole in kilograms, given that 1 cm^3 of iron has approximately 8 g mass. [6]

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 6 Marks]



Step 1: Listing Dimensions & Volume Formula [1.5 Marks]
Lower Cylinder: $R = \frac{24}{2} = 12$ cm, $H = 220$ cm. Upper Cylinder: $r = 8$ cm, $h = 60$ cm.

Formula / Theorem Used: $V_{\text{total}} = \pi R^2 H + \pi r^2 h = \pi(R^2 H + r^2 h)$

Step 2: (i) Total Volume Calculation ($\pi = 3.14$) [2.5 Marks]

$$V_{\text{total}} = 3.14 \times (12^2 \times 220 + 8^2 \times 60) = 3.14 \times (144 \times 220 + 64 \times 60)$$

$$= 3.14 \times (31680 + 3840) = 3.14 \times 35520 = \boxed{\mathbf{111532.8}\ \text{cm}^3}$$

Step 3: (ii) Mass in Kilograms [2 Marks]
Mass of 1 cm^3 iron = 8 g:

$$\text{Total Mass} = 111532.8 \times 8\ \text{g} = 892262.4\ \text{g}$$

$$= \frac{892262.4}{1000}\ \text{kg} = \boxed{\mathbf{892.2624}\ \text{kg}\ (\approx \mathbf{892.26}\ \text{kg})}$$

★★★ END OF SOLUTIONS · MATHS MANTHAN MEERUT ★★★